

# Ontology based Machine Learning Approach for Facial Skincare Products Recommendation

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**Abstract**— The need to preserve facial skin health and improve attractiveness has become more widespread in the modern world. There has been competition among skincare companies to research and create novel products. The dynamic skincare market creates a wide range of skincare products. Therefore, selecting the best skincare products suitable for a consumer's skin type and condition can be quite challenging. Skin impairments could worsen if products contain ingredients inappropriate for the user's skin type. This study proposes a unique system architecture for recommending facial skincare products that combine ontological and machine-learning benefits. The ML engine is developed for facial skin condition identification, which uses a CNN model to identify acne severity based on the user's facial image. An ontology was constructed using the Protégé ontology editor, which included hierarchical relationships between user profiles, skincare information, and skincare product information. The semantic similarity between these concepts was mapped by the Protégé tool for the skincare product recommendation engine. The system UI takes inputs for customization and recommendation of facial care products based on key factors such as the user's skin type, concerns, acne severity level, and allergy ingredients. Users can provide feedback and ratings for the products recommended. The developed system had an accuracy of 87.5% based on a survey conducted with 24 participants who tested the system. Therefore, the proposed ontology-based ML approach is effective and accurate for facial skincare product recommendations.

**Keywords**—Ontology, ML, CNN, facial skincare products

## I. INTRODUCTION

At present, skincare is a global interest shared by both men and women. The skincare industry has seen an increase in demand and popularity due to the rising investments made in the research and development of novel skincare products. In 2021, the skincare industry was valued at USD 130.50 billion [1]. The increased creation of facial skincare products has been sparked by people's growing knowledge of the value of preserving facial skin health while enhancing their appearance and preventing skin disorders [2]. The extensive and dynamic nature of the skincare market has created a dilemma for consumers in selecting skin care products that are suitable for their skin impairments and skin types. As a result, extensive research has been done to personalize and suggest facial care products based on customer feedback and skin diagnosis.

Even if facial acne and other treatable skin conditions like pigmentation, wrinkles, and dullness are not serious disease conditions, they can have a significant psychological impact and interfere with a person's social life. Consequently, maintaining the appearance of healthy facial skin helps boost a person's self-confidence. For consumers,

choosing facial skincare products is not an easy undertaking with the availability of a wide variety of specifications in cosmetics, whether they purchase goods from pharmacies, department stores, or online. They seek out recommendations from friends, doctors, or their favorite bloggers and frequently buy expensive products that do not deliver the desired results [3].

According to the research [4], [5], [6], and [7], AI has demonstrated significant potential for identifying skin type and skin problems and recommending skin care products. Among the many other AI technologies that have produced skin diagnosis tools and skincare product recommendation systems are ML, Expert Systems (ES), ontology, Convolutional Neural Networks (CNNs), and Natural Language Processing (NLP).

In this study, a hybrid strategy is put forth that combines the advantages of machine learning and ontology. The system analyzes skin conditions and skin types and makes suggestions for skincare products using methods of ML and ontological modeling.

This system has been developed on the premise that facial skincare product recommendations can be made as an AI solution, using ML to identify skin conditions and ontology to represent the skincare domain and user profiles. The user's information is taken for product personalization via the User Interface (UI), such as facial image, skin type, skin conditions, and allergy ingredients. The acne severity of a person will be predicted using the facial image input provided to the system. To predict acne severity, the ACNE04 [8] dataset was customized and trained on CNNs to identify the features and patterns of acne severity classes: mild, moderate, and severe. Using the Protégé open-source ontology editor, user profiles and skincare information were mapped as hierarchical classes and relationships between them as object and data properties, allowing the inferencing of new knowledge that is not capable of traditional databases. Then customized skincare products were recommended using the semantic similarity of the user information, facial skincare concepts, and product information.

The rest of the paper is structured as follows: In Section II, the skincare domain literature is critically reviewed, highlighting the technologies adopted, state-of-the-art solutions, potentials, and limitations of skin care product recommendation systems. Section III discusses system development, focusing mainly on ontology modelling and ML model development using CNNs. Section IV explains in detail the thorough evaluation procedure of the developed system, which includes getting feedback from experts and users of the system and its analysis. In Section V,

conclusions obtained from the research are highlighted, and recommendations for prospective future works are made.

## II. BACKGROUND

As the largest organ, the skin is faced with multiple environmental hazards daily, and with chronological aging, skin problems become more evident when compared to well-maintained, younger-looking skin. Research has shown that everyday skincare routines potentially have a significant long-term impact on a person's overall complexion quality [3], [9].

Skin aging is the main skin impairment people face with time. It can be classified as extrinsic or intrinsic aging. External factors are responsible for 80% of the apparent indicators of skin aging, such as UV radiation, pollution, and lifestyle decisions like smoking, sleeping positions, diet, and daily skincare practices, while only 20% are caused by intrinsic factors, such as genetics, general health, and stress levels. Some overt signs of intrinsic aging are thinning of the stratum corneum, atrophy, fine wrinkles, and dryness, while some of the visible signs of extrinsic aging are thickening of the epidermal layer, uneven pigmentation, dullness, dryness, creases, and laxity, which are the main reasons people seek dermatological treatments [3].

Acne vulgaris is a skin condition that affects a person's physiological conditions, self-esteem, and confidence. Acne stigmatizes both adults and teenagers. This skin condition develops when oil and dead skin cells clog hair follicles. It causes whiteheads, blackheads, or pimples, leading to widespread nodulocystic lesions with the potential for scarring. Post-inflammatory hyperpigmentation is more likely to occur in people with darker skin types. It causes much mental anguish for people suffering from acne [3], [10]. Early diagnosis, together with accurate, continuous self-tracking, would benefit controlling and treating this illness. Since the majority of consumers tend to treat their acne on their own, helping them select a successful skincare plan will improve their overall physical appearance and increase their contentment. Acne treatments are based on how severe the condition is. Over-the-counter (OTC) products can be used to manage mild to moderate acne severity. In order to treat severe to very severe cases, dermatologist treatment is required. Therefore, based on literature as well as skincare experts' guidance, users with severe and very severe acne cases are directed by the proposed recommender system to seek dermatological treatments. To treat mild and moderate acne conditions, OTC products such as gels, soaps, creams, and lotions that consist of ingredients such as salicylic acid, benzoyl peroxide, and sulfur have been recommended to be applied to the skin [3], [11].

There are many state-of-the-art skincare product recommendation systems, such as rule-based, ML-based, Content-Based Filtering (CBF), Collaborative Filtering (CF), etc. Rules-based expert systems that make skincare product recommendations imitate the work of skincare experts who interpret information from humans to computers. Users are queried by the inference engine, and their responses are recorded for analysis. The user's evaluation of their skin condition is then examined utilizing forward and backward chaining methods, and skin care products are suggested [6], [12].

The study by Pornchai Nopparatkiat and colleagues proposed an expert system-based method for the treatment of skin problems using Thai herbal recipes and Thai herbal cosmetics, adopting Hypertext Preprocessor (PHP) and MySQL database technologies. The created expert system consists of the primary knowledge-based tools that diagnose skin issues such as acne, melasma, freckles, wrinkles, and uneven skin tone. In this recommendation system, the knowledge base was limited to these skin impairments that were in the database and suggested treatments for those issues, including safeguarding and preserving skin health. Also, another major drawback of the system was that the accessibility of Thai herbal medicines and cosmetics was limited to Thai natives and not readily available internationally [6].

Abdullah and Basari developed an ES that provided users with information and consultations to identify skin types [13]. It recommends treatment solutions for the skin type and helps avoid the formation of any serious skin disorders. The database system used for the system's development is MySQL, and the programming language used is PHP. The advantages of the Skincare Routine Expert System were that it enabled users to quickly determine the type of skin they have and receive recommendations for the best treatments. One of the major limitations of this ES is the creation of a set of rules given the abundance of information regarding the signs and symptoms of the acquired disease. Due to this, a finite number of rules utilizing database technologies cannot express all information or facts about skincare disorders and treatment solutions.

In rules-based expert systems, the user's assessment of their skin's state is considered when analyzing using forward and backward chaining techniques. However, some dermatologists have had difficulty making a diagnosis with the naked eye because several skin illnesses are identical [6], [12]. Also, in some scenarios, only a limited number of products stored in the system's database are recommended, and due to the dynamic and extensive nature of the skincare domain, a limited number of ES rules cannot recommend skincare products for all skin concerns, and ES rules are not updated regularly. The main advantage is that ES provides explanations for the product suggestions so that users have a comprehensive idea of the concerns and treatment products.

The study conducted by Ting-Yu Lin et al. [14] identified if the person's skin type was oily, dry, or neutral, and classified the severity of acne using ML classification technology, and provided recommendations for skincare products. Here, the facial image of the user was captured using Logitech's C310 camera, and then the Open CV facial recognition model was used to detect facial regions in the image. Using the regions of interest captured, which were the cheeks, forehead, and chin, the skin type was classified, and the severity of acne level was determined by the proportion of acne areas detected over the ROI image. Then, the skincare products that are stored in the database were recommended through the UI based on the predicted skin type and acne severity level (mild, moderate, and severe). The overall work provides a novel business model composed of a finger vein identification system, a skincare product recommendation system, and an electronic payment system.

The capacity to explain product recommendations, not considering the user's input about additional skin problems for personalization and recommending skincare products that

are merely kept in the system database are the main shortcomings of ML-based systems. A major improvement is that the user's skin type and acne severity level were considered in product recommendations.

CBF and CF algorithms are used mostly in recommender systems, as per the literature [15], [16]. Based on the similarity of the users' grading of prior items or the ratings provided by the individuals, collaborative filtering makes recommendations. Collaborative filtering is a widely used technique. The drawback of collaborative-based filtering is the cold start problem. Making recommendations is challenging when the users or the items are new. Also, another drawback is that it is susceptible to fraud user attacks, in which several fictitious user profiles are entered into the system to affect the user recommendations, potentially having a significant negative impact on the systems. With more people and items, collaborative filtering performance suffers from scaling issues. As a result, the collaborative filtering algorithm needs to be modified frequently over time [16]. Based on item descriptions and user preferences, content-based filtering recommends suitable items to users. Active users can supply reliable information even if they are not logged in, so content-based filtering benefits them. The limitation of content-based filtering is that it cannot recommend suitable products when the content analyzed for an item does not contain adequate information.

Thus, the literature review has revealed many limitations in the area of skin care product recommendations. More importantly, many researchers have recognized the need for skincare product recommendation systems given the rate at which new skincare products are arriving on the online market. Therefore, we define our research problem as the fact that the current approaches to skincare product recommendations are rather inefficient and inaccurate when the skin care domain is changing rapidly, and consumers find it difficult to select a suitable facial care product that suits their skin condition and skin type.

### III. METHODOLOGY

The ontology-based ML framework for facial care product recommendation utilizes a hybrid filtering strategy, combining ML and ontology technologies. The system consists of three layers, as shown in Fig. 1. The data gathering layer is the first layer, which includes the data collection module and all the information resources. This is used to accumulate all the facial skin-related important concepts and skincare product-related information needed for precise, customized product recommendations. The second layer consists of the primary functional components of the system. They are the ML engine for acne severity prediction, the ontological data model, and the recommendation engine. The user application layer enables user interaction with the framework and is the third and last layer. It allows users to make selections, enter their skin details for personalization, and provide feedback on the recommended products.

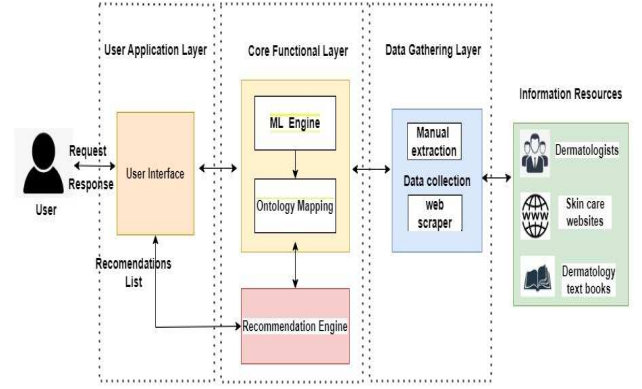


Fig. 1. System architecture for skincare product recommender system.

The main system architecture depends on the conceptualization of the skincare domain and the ontological mapping of skincare concepts, product information, and user profile information. The skincare domain concepts are dynamic and vast, and to build the vocabulary of terms for ontological mapping, expert support and guidance was sought by interviewing dermatologists and health professionals. Also, a survey aimed to understand consumers' skincare habits was further conducted, involving 11 male and ten female participants aged 21-30. Participants were asked about their skincare routine, purchasing habits, and preferred skincare brands for product recommendation. Then data collected was preprocessed from an Excel sheet is organized for easy creation of ontology classes, subclasses, and instances.

Protégé is an open-source software for building and managing ontologies and knowledge bases. In the proposed recommender system, Protégé was used as the primary development environment for ontology modelling, utilizing Pellet reasoner for verifying consistency and obtaining logical conclusions. System development involved modeling three ontologies which were: skincare concepts ontology, skincare product information ontology, and user profile ontology, which was later merged to form a single skincare domain ontology. As depicted in Fig. 2, Fig. 3 and Fig. 4 skin information ontology, user profile ontology and skincare product ontology respectively.

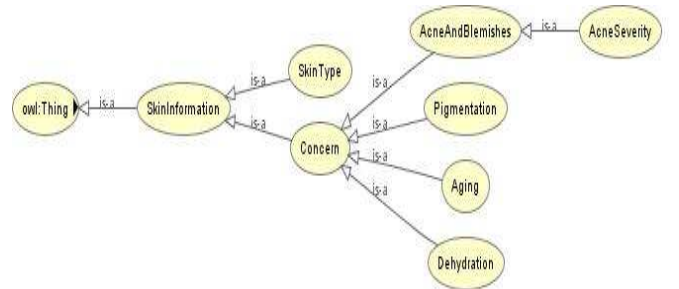


Fig. 2. Illustration of the skin information ontology.



Fig. 3. Illustration of the person ontology.

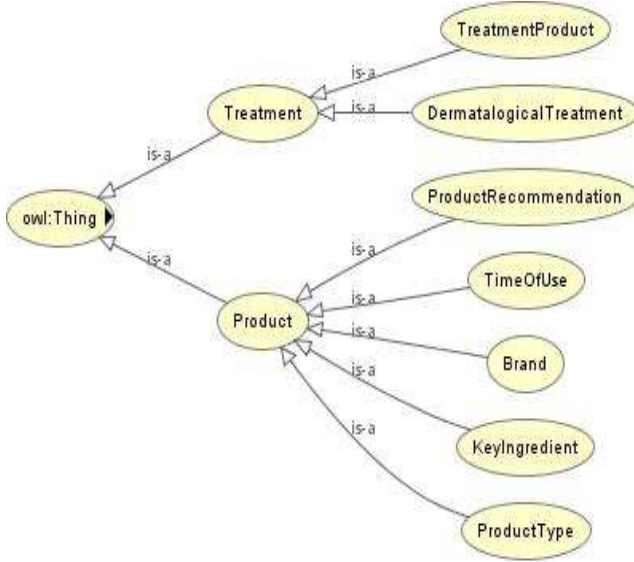


Fig. 4. OWLViz representation of the skincare product class hierarchy.

In ontology construction, class hierarchies were defined using the top-down approach. That is, to define the general skin care concepts first, and then, moving down the class hierarchy, specialized concepts are stated.

In the next step of ontology construction, properties are defined. The relationship between members of a class is known as the properties of an ontology. Data properties and object properties are the two basic types of properties. The object properties serve as a representation of the binary relationships that exist between members of the classes [17]. For example, the suitableFor object property is defined between the TreatmentProduct (domain) class and the SkinType (range) class. It shows the relationship or feature in skin care products that is suitable for certain skin types, as shown in Table I. Table I shows some of the object property relationships defined when constructing the ontology, with their domain and range and an example instance for each class.

In the other property type, an individual is linked to a data literal via a data property, such as defined for Person class hasAge, hasGender as shown in Table II.

TABLE I. OBJECT PROPERTIES DEFINED IN THE ONTOLOGY.

| Object Property          | Domain                                  | Range                                      | Description                          |
|--------------------------|-----------------------------------------|--------------------------------------------|--------------------------------------|
| suitableFor              | Treatment Product (AcneControlCleanser) | SkinType (OilySkin)                        | Suitable skin type for the product   |
| hasKeyIngredient         | Treatment Product (AcneControlCleanser) | KeyIngredient (SalicylicAcid, Niacinamide) | Key Ingredients of treatment product |
| hasProductRecommendation | Treatment Product (AcneControlCleanser) | ProductRecommendation (productreco)        | Rating and recommendation            |

|  |               |               |                                     |
|--|---------------|---------------|-------------------------------------|
|  | rolCleanser ) | mmendation 8) | endatio n given as feedback by user |
|--|---------------|---------------|-------------------------------------|

TABLE II. DATA TYPE PROPERTIES DEFINED IN THE ONTOLOGY.

| Data Property | Domain                | Range      | Description                            |
|---------------|-----------------------|------------|----------------------------------------|
| hasAge        | Person                | xsd:int    | User age                               |
| hasGender     | Person                | xsd:string | Gender of the user                     |
| hasRating     | ProductRecommendation | xsd:int    | Rating given for a product by the user |

In ML engine development, a CNN model to identify three different types of acne severity classes using the custom-built ACNE04 dataset by Xiaoping Wu and his colleagues [8] was proposed in the study. Since the data was imbalanced, severe and very severe classes were merged to form one prediction class as severe. Also, to improve accuracy and avoid overfitting, the dataset was customized with images selected from the skincare website DermNet (dermanetz.org), and newly labeled images were put into each class. The customized data was used to train and construct the CNN model.

Then the data was preprocessed using scaling and data augmentation techniques. A few of the augmentation techniques used in model development were horizontal and vertical flips, rotations, color jitters, and translations. In the next stage, a CNN model was created using the Keras Sequential API by adding layers sequentially to the model architecture. The first layer is a convolution layer (Conv2D) with 16 filters and a stride of one to extract local features from the input images. The model is built of two more layers of Conv2D with filters of 32 and 16, respectively. Following the Conv2D layer, Max Pooling Layers are applied to minimize the feature maps' spatial dimensions and capture key features. This layer down-samples by choosing the highest value from a pool of values in a local region. It also helps in preventing overfitting issues.

- A flatten layer (Flatten) was inserted after the convolutional and max pooling layers to transform the 3D feature maps into a 1D feature vector that can then be utilized as input to fully connected layers.
- The next addition is a fully connected dense layer (Dense) with 512 units with the ReLU activation function. This layer uses a matrix multiplication operation and element-wise activation to extract global patterns and representations from the feature vector that has been flattened.
- After the dense layer, a dropout layer (Dropout) with a rate of 0.5 is inserted to avoid overfitting. Dropout assists in regularizing the model and preventing an

overdependence on certain neurons by setting a random percentage of input units to 0 during training.

Since this model is developed for 3-class classification, the output layer is a dense layer of 3 units with a softmax activation function. When classifying images, each class's distribution probability is output by the neural network.

Next, the Python Tkinter standard library was used to develop the GUI of the recommender system. Since it is lightweight and cross-platform, the built desktop application can be used across multiple operating systems, such as Windows and MacOS. The client can interact via the built-in UI and enter personal details as well as skin-related information. The data captured, therefore, by the UI is the username, age, gender, facial image, acne severity, skin type, product type, skin concern, and any allergy ingredients.

Finally, in the recommendation engine development, the main inputs from the user are queried against the skincare domain knowledge base. The output from the ML engine is also further enhanced by querying against the skincare domain knowledge base when recommending products using the recommender engine. The Owlready2 Python library has been used, which provides a convenient way to manipulate ontology classes, instances, and properties similar to handling regular Python objects. This library was used to query the inputs taken via the UI with the OWL ontology. Fig. 5 and Fig. 6 show the various UIs of the proposed recommendation system.

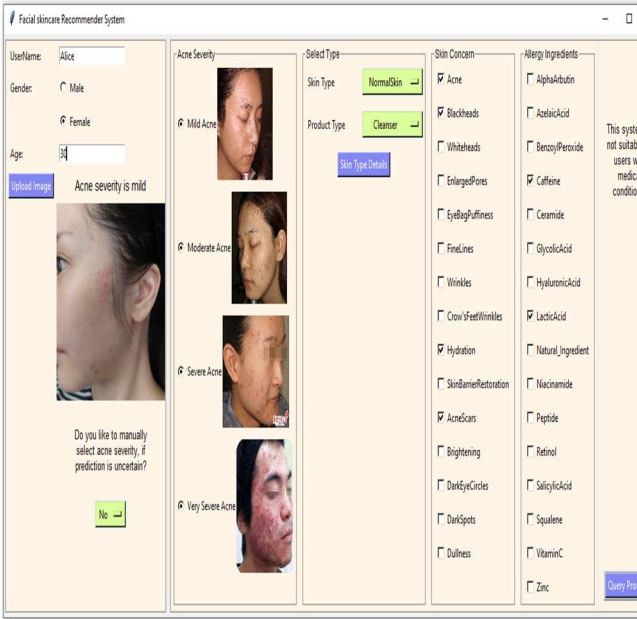


Fig. 5. Screenshot of the main screen of facial skincare recommender system.

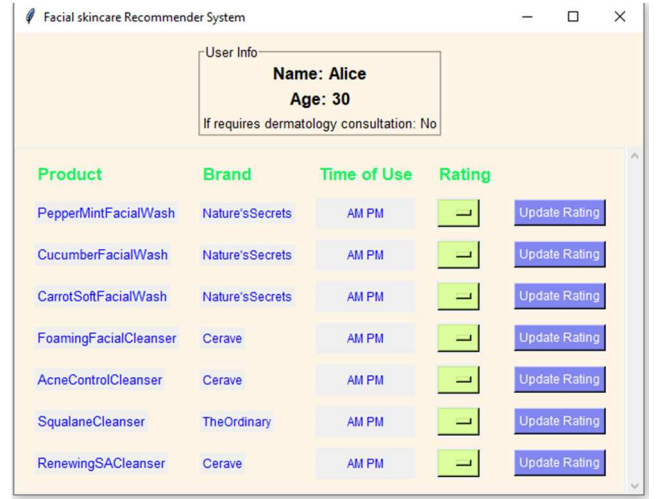


Fig. 6. Screenshot of the product recommendation screen.

#### IV. EVALUATION

The objectives of this study were to design and develop an ontology-based machine learning approach for facial skincare product recommendation and evaluate the prototype of the system. Experimental results were collected based on the survey conducted. This study used 24 test subjects (12 males and 12 females). A questionnaire was prepared to determine consumer satisfaction and the quality of the suggested skincare products for the users. Five-point Likert-scale survey questions were asked.

From the analysis of the results of the survey conducted, 87.5% of the participants said that the products were accurate, suitable for their skin type and skin concerns, and personalized according to their skincare needs. Also, 91.7% of the users stated that the user interface was easy to use and not complicated. Moreover, 91.6% of the users were satisfied with the overall functionality and user-friendliness of the facial skincare recommender system. As new facial care products were introduced to the users by the system, depending on the results, 66.7% of participants stated that they were familiar with the products recommended, and 95.9% affirmed that the recommended products helped them discover new products suitable for their skin requirements.

Standard ML evaluation techniques were used in model performance evaluations. Thus, the model had a precision, recall, accuracy, and F1-score of 78.2%, 75.1%, 77.5%, and 76.6%, respectively. It performs well against the state-of-the-art prediction models, and its performance will be further enhanced by the ontology modeling of the skincare domain for product recommendations.

#### V. CONCLUSION

With the dynamic and constantly expanding skincare industry, users are presented with a myriad of available facial care products. Selecting the skin care products that best suit the skin type and conditions is vital; otherwise, allergic reactions might occur with products that contain ingredients that are incompatible with the user's skin type. The existing skincare product recommendation systems fail to impart thorough knowledge about skincare product details and are not personalized to address individual users' skincare needs. This study presented a novel hybrid approach to providing recommendations based on ML and ontology. Key features include an easy-to-use UI that helps to select skin care

products based on skin type, concern, acne conditions, and product type while filtering out any allergy ingredients. It provides users with access to a wide variety of skincare products in a concise time frame as well as acne severity prediction for their facial skin. Acne severity was selected as the skin condition because, based on the literature, the majority of people suffer from acne vulgaris. Also, the acne severity level is a key parameter examined when dermatologists recommend facial care products. This ontology-based recommender system is knowledge-based and employs ontologies to map knowledge about skincare concepts, facial care products, and user profiles in the recommendation process.

Furthermore, the product suggestions recommended promptly consist of the following information: product name, brand name of the product, time of use of the product, and product rating. Also, the system provides user guidance on seeking dermatological assistance based on the severity of the acne. These features can be considered distinctive features of the system in comparison to other recommender systems. The recommendation of our system's skin care products is targeted at young adults and adults in the age range of 16 to 60. The duration of usage of recommended skincare products varies widely depending on various factors, including the individual's skin type, the specific concerns being addressed, and the efficacy of the products being used. Before anticipating noticeable changes, it is generally advised to follow a skincare regimen consistently for at least four to six weeks.

Based on the results of the survey conducted, more than 87.5% of participants approved of the suggested skincare products, and the results matched their skincare requirements. The evaluation results show that this machine learning and ontology-based facial skincare recommender system is effective and accurate for skincare product recommendations. Thus, the aim of this research has been successfully accomplished.

Additionally, based on the literature, dermatologists sometimes face difficulty identifying different acne types due to their overlapping features. So, further development of the ML module can include identification of skin conditions such as acne types, rosacea, and eczema, which helps the user recognize their skin conditions and receive necessary dermatological treatments as well, which can further improve the performance of the system.

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